

User Manual

FIT AR Navigation App (FITARNA)

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1. Introduction

FITARNA is an augmented reality navigation application designed for use inside the Florida Tech library. The app's purpose is to help users navigate the library more easily while also providing an interactive experience through augmented reality visuals and informational pop-ups.

Using FITARNA, users can select different floors and destinations within the library and receive visual guidance to help them reach their chosen location. In addition to navigation, the app enhances the user experience by displaying informational AR pop-ups that share the library's history, fun facts, and other helpful details.

FITARNA is intended to make the library easier to explore for students, faculty, staff, and visitors. Whether a user wants to quickly find a destination or casually explore the building, the application provides a more engaging and intuitive way to navigate the space.

1.1. Purpose of This Document

This user manual provides guidance for installing, configuring, and using the FITARNA application. It is intended to serve as a reference for all users of the Evans Library at Florida Institute of Technology. Sections cover device compatibility, installation procedures, operational modes, in-app AR features, and troubleshooting steps.

1.2. Intended Audience

FITARNA is designed for anyone who enters or wishes to explore the Evans Library at Florida Tech. This includes:

- Students enrolled at Florida Institute of Technology
- Faculty and staff members using the library's resources
- Visitors, prospective students, and guests touring the campus
- Library personnel seeking to understand the app from a user's perspective

No prior experience with augmented reality applications is required. The app is designed to be intuitive and accessible to users of all technical backgrounds.

If additional help is needed after downloading or using the app, users may contact the FITARNA development team using the email addresses listed at the beginning of this manual.

2. Downloading the App

2.1. Compatible Systems

At launch, FITARNA is available only on Android devices. The app is primarily designed for Android smartphones, but it may also work on compatible Android tablets.

Because FITARNA uses augmented reality, the device must support the hardware and software features required for AR functionality. For the best experience, users should use a modern Android device with a working camera, motion tracking capability, and up-to-date system software.

Support for iPhone and other platforms may be considered in future versions of the application.

2.2. How to Install

FITARNA is installed by scanning a provided QR code that links directly to the application's .apk download file.



To install the app:

- Open the camera app or a QR code scanner on your Android device.
- Scan the QR code provided for FITARNA.
- Tap the download link for the .apk file when it appears.
- Follow the on-screen prompts to download and install the application.

Some Android devices may block the installation of applications downloaded outside of the default app marketplace. If this happens, the device may prompt the user to allow installation from unknown or external sources. If prompted, the user should temporarily enable this permission in their device settings to complete the installation.

After the installation finishes, the FITARNA app icon should appear on the device. Open the application and grant any required permissions, such as camera access, so the augmented reality features function correctly. Without these permissions, the app could not function as intended.

It is good practice to re-disable the Unknown Sources permission after FITARNA has been successfully installed to maintain the device's default security settings.

2.3. Updates

FITARNA may receive updates over time to improve performance, fix bugs, enhance navigation accuracy, and add new features.

When a new version of the application is released, users may be asked to install a newer .apk file via an updated QR code or download link. It is recommended that users install the latest available version to ensure the best performance and the most reliable user experience. Using an outdated version may result in reduced performance and missing improvements.

To update the app, scan the new QR code or visit the updated download link and follow the same installation steps as the original install. The new version will replace the existing one on the device.

2.4. Getting Started

Once FITARNA is installed/updated, the user can open the application on their device as they would any other mobile app. Users should already be physically inside the library when starting the app.

FITARNA uses pre-scanned spatial data of the library's interior to understand the environment. This means the AR tracking and navigation features are calibrated specifically for the library's physical space. Using the app outside of the library will result in inaccurate or non-functional AR behavior.

Once the user is in the library and the app is open, users can optionally take a moment to look around with their phone camera to help the app localize. Users can then choose between the two app modes on the app's home screen. Users who want to go directly to a room or area may select Navigation Mode, while users who want to experience the informational tour may select Tour Mode.

3. App Modes

3.1. Tour Mode

Tour Mode allows users to explore the library freely while viewing augmented reality pop-ups placed throughout the environment. This mode can be activated by selecting the Tour Mode button on the FITARNA home screen.

When Tour Mode is active, users can move through the library at their own pace and view informational content connected to different areas of the building. These pop-ups may include historical details, trivia, or interesting facts about the library and its spaces.

Tour Mode is intended to create a more engaging and educational experience for users who want to learn more about the library rather than just visit a single destination. It is especially useful for first-time visitors, prospective students, and anyone interested in exploring the space interactively.

Tour Mode does not provide turn-by-turn navigation. Users are free to move through any floor or area of the library at will. Pop-ups will appear automatically when the user is near a point of interest, and the device's camera detects the associated environmental markers.

To exit Tour Mode at any time, use the back button or the in-app menu to return to the home screen.

3.2. Navigation Mode

Navigation Mode helps users travel directly to a selected destination within the library. This mode can be activated by selecting Navigation Mode from the FITARNA home screen.

Once Navigation Mode is opened, the user can choose a destination within the library. Available destinations may include specific floor levels and study areas; reference and circulation desks; computer labs and printing stations; meeting rooms and group study spaces; restrooms; elevators; and other facilities.

FITARNA will then guide the user using augmented reality navigation cues and visual guidance to help them reach that location.

This mode is useful for users who need help locating a specific room, service area, study space, or other point of interest inside the building. By offering real-time guided navigation, FITARNA reduces confusion and makes indoor wayfinding easier and more accessible.

If the AR cues disappear during navigation, slow down and hold the camera steady for a moment. The app may need a few seconds to re-localize before navigation cues resume. See Section 5.2 for more information on localization.

4. App Pop-ups

FITARNA includes in-app augmented reality pop-ups that appear as virtual information boxes within the user's view. These pop-ups are designed to provide useful and engaging information as users navigate the library. They are among the app's main features that distinguish FITARNA from other navigation applications.

The pop-ups are intended to be easy to notice, quick to read, and visually clear. Their purpose is to share information about the library in a format that supports the user experience without becoming distracting. They appear in the augmented reality view to help the user feel more connected to their surroundings.

4.1. Types of Pop-up Content

Depending on the location and app mode, these pop-ups may display:

- Historical information about the library
- Fun facts or trivia
- General points of interest
- Helpful context related to the user's surroundings

These features help make FITARNA more than just a navigation tool. They also allow the app to serve as an interactive guide to the library environment.

4.2. Interacting with Pop-ups

Pop-ups appear automatically when FITARNA detects that a user is near an associated location and their camera is oriented in the appropriate direction. No manual interaction is required to trigger them.

Users can read a pop-up and then continue walking. The panel will gradually fade from view as they move away from the associated location. If a user wishes to view a pop-up again, they can return to that area and point the camera at the same location.

In certain cases, pop-ups may contain additional details accessible by tapping the panel. Not all pop-ups support interaction. Pop-ups are visible only when the camera is reasonably stable. Rapid movement or camera shake may cause pop-ups to flicker or disappear temporarily. This is normal behavior related to the AR tracking system.

5. Troubleshooting

If FITARNA does not function as expected, several factors may be at play.

5.1. Permissions

If the user starts the FITARNA app and sees a black screen after selecting either mode, the likely cause is a permissions issue. Augmented reality features depend on camera access, so denying this permission will prevent the app from displaying the user's view or any AR content.

To fix this issue:

- Close the FITARNA application completely.
- Open the device's Settings application.
- Navigate to Apps or Application Manager and find FITARNA.
- Select Permissions and ensure that Camera access is set to Allow.
- Reopen FITARNA.

On some Android versions, Motion or Sensors permissions may also be required for full AR functionality. Check all available permissions in the FITARNA app settings if issues persist after enabling camera access.

5.2. Localizing

If tracking or AR content appears inaccurate, the issue is likely due to localization. The FITARNA app uses the camera and saves library scans to localize. If the user's settings in the library differ from those the dev team scanned during development, the app is likely to be confused about the user's position. Additionally, if the user is moving too quickly or isn't displaying their position and movement to the camera, localization is bound to be lost.

A way to aid localization is to point the camera at close-up, simple objects. Pillars and walls are effective at quickly reorienting the user. When moving, keep the camera at a consistent height and point it in the direction of travel.

If localization problems persist after trying these steps, the user should restart the FITARNA application and try again in a well-lit area of the Evans Library.

5.3. Performance / Lag

If the FITARNA app feels slow, delayed, or unresponsive while being in use, the issue is likely related to device performance and the demands of AR processing. FITARNA relies on the camera, motion tracking, rendering, and real-time pathfinding. As a result, some devices may experience reduced performance, especially older ones.

A simple fix to performance and lag issues is to simply close as many background applications as possible and restart FITARNA. Users should also try to ensure

their device is not in a low-power state or low-power mode. Battery-saving features will likely reduce performance and interfere with intended app behavior.

If performance issues and lag persist after implementing those fixes, the lower performance may simply be due to device limitations.

5.4. Installation Issues

If FITARNA does not install correctly, the cause is likely the user's device blocking applications downloaded outside the app marketplace. FITARNA is currently installed via a QR code or an .apk file, so some Android devices may automatically prevent installation.

If this issue occurs, the user should open the device settings and enable permission to install apps from external sources. After this, try installing again. After installation, the user can turn off the permission once again.

If the QR code for FITARNA does not initiate the download, the issue could be the internet connection or the QR code scanner. Try ensuring internet connection, scanning the code again, or using a different QR scanning app.

5.5. Camera and AR Issues

If the camera feed appears blurry, distorted, or otherwise abnormal, check the following:

- Ensure the camera lens is clean and free of smudges or obstructions.
- Restart the FITARNA application to reinitialize the camera session.
- If other camera applications also exhibit unusual behavior, the issue may be a hardware or system-level problem with the device itself.

If AR overlays such as navigation cues or pop-ups appear but are visibly misaligned with the physical environment, this is typically a localization issue.

Refer to Section 5.2 for guidance on improving localization accuracy

6. Frequently Asked Questions

Q: Does FITARNA require an internet connection to use?

A: No. FITARNA operates entirely offline once installed. All map data, spatial scans, and pop-up content are stored locally within the application. An internet connection is only needed during installation or when downloading updates.

Q: Can I use FITARNA outside of the Evans Library?

A: FITARNA is calibrated specifically for the interior of the Evans Library at Florida Institute of Technology. Using the app outside of the library will result in inaccurate or non-functional AR behavior. The app is not intended for general-purpose indoor navigation.

Q: Does FITARNA work on multiple floors of the library?

A: Yes. FITARNA supports navigation and pop-up content across multiple floors of the Evans Library. When using Navigation Mode, users may select destinations on different floors, and the app will guide them accordingly, including to stairways and elevators as appropriate.

Q: What should I do if a pop-up does not appear where I expect it?

A: Pop-up placement depends on accurate localization. If a pop-up is not appearing as expected, try re-localizing by pointing the camera at nearby pillars or walls and walking slowly. If the problem persists, restarting the application may resolve it.

Q: Can I use FITARNA with headphones?

A: FITARNA is primarily a visual application and does not currently include audio narration or audio cues. Headphone use does not affect the app's functionality.